Cantera vs NN | Test IC 57 | NRMSE (util.py)

T0 = 1412.0 K P0 = 4.083 atm phi0 = 0.677 | Repaired self-fed, 10 ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Training	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	**10k**	0.0428	0.666	8.62	0.853	0.176	0.956	0.0319	1.51	4.49	0	Complete
2 P-only DeepONet10	**10k**	0.0101	0.104	0.933	0.986	0.0348	0.33	0.0267	0.132	1.47	0	Complete
2 P-only DeepONet10	**long**	2.84e-03	0.323	0.857	0.952	5.85e-03	0.233	3.52e-03	0.0448	0.0253	0	Complete
3 ANN10 + P-only DeepONet10	**10k**	5.19e-03	0.0455	0.906	1.27	0.0366	0.352	0.0232	0.0753	1.05	0	Complete
3 ANN10 + P-only DeepONet10	**long**	3.19e-03	0.298	0.87	1.07	0.0111	0.253	5.68e-03	0.0357	0.0136	0	Complete
4 T/P DeepONet10	**10k**	3.80e-03	0.0682	0.0852	0.0645	2.64e-03	0.0271	4.83e-03	0.0327	0.023	0	Complete
4 T/P DeepONet10	**long**	0.0326	0.181	0.466	0.189	0.0562	0.681	0.0453	0.47	0.442	0	Complete
5 ANN10 + T/P DeepONet10	**10k**	4.76e-03	0.0786	0.0856	0.0623	1.90e-03	0.045	6.40e-03	0.0294	0.0218	0	Complete
5 ANN10 + T/P DeepONet10	**long**	0.0188	0.081	0.233	0.0214	0.0372	0.455	0.0278	0.281	0.3	0	Complete
6 T/P/ϕ DeepONet10	**10k**	8.28e-03	0.0376	0.0669	0.036	0.03	0.288	0.0133	0.265	0.371	0	Complete
6 T/P/ϕ DeepONet10	**long**	5.07e-03	0.0825	0.107	0.0449	0.0148	0.143	8.31e-03	0.15	0.115	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**10k**	3.84e-03	0.0728	0.087	0.0721	0.0256	0.213	7.18e-03	0.213	0.325	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**long**	2.88e-03	0.0541	0.0529	0.0535	0.0142	0.128	5.34e-03	0.115	0.134	0	Complete

Exact util.py defaults: norm="rms", reference A=Cantera. Every time sample has equal weight; entries are dimensionless ratios, without x100.
N/A: zero Cantera RMS with a nonzero prediction. Constant nonzero references, including N2, are valid. —: prediction stopped early; full-horizon NRMSE is unavailable.
ANN stays at 10k in every mixed team. Long DeepONets stopped at their wall-time caps; no verified loss plateau.
All columns compare the NN curve with the same IC Cantera curve. First graph row has only the ANN10k result.